

Automatic Prostate Segmentation on MR Images using Enhanced Holistically-Nested Networks

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Abstract—Magnetic resonance(MR) imaging has shown to be succeed in detecting and visualizing the prostate location. The accurate segmentation of the prostate gland from MR images is necessary for clinical applications. However, the segmentation of prostate is also a challenging task because of the shape of prostate varies significantly and the inhomogeneous intensity distributions in different scans. In this paper, we present an automatic deep learning method for prostate MR images segmentation using the enhanced holistically-nested framework. The network Holistically-Nested Networks(HNN) was first proposed as an image-to-image solution to extract object edges and boundaries visually. We modify HNN via putting additional skip connections from later stages to early stages in order to combine both low-level features and high-level features. The deeper framework exploits multi-level and multi-scale information for the image-to-image prediction in a holistic manner. Experimental evaluation demonstrates that significant segmentation accuracy has been achieved by our proposed enhanced holistically-nested networks compared to other deep learning approaches.

I. INTRODUCTION

Researchers in the relevant institutions have found that the prostate cancer is already the second major disease of cancer death of males in the United States [1], [2]. The clinical diagnosis of prostate cancer requires segment the prostate gland to different regions. The Computer-aided detection(CAD) has been proposed for improving the accuracy of prostate cancer treatment [3], [4]. The segmentation of prostate volume is often used as the first step in CAD algorithms. Magnetic resonance(MR) imaging is a principal detection of prostate shape and location due to its high soft-tissue and spatial resolution. The prostate MR image segmentation has attracted much research interest [5]. Recently, prostate MR images are mainly segmented by radiologists with rich experience in medical treatment. The manual annotation is time-consuming and requires high degree of skills. Moreover, prostate segmentation is also a challenging task because MR imaging are often influenced by artefacts and inhomogeneous intensity distribution. Fig. 1 shows us three different forms of prostate MR slices.

Convolutional neural networks(CNNs) are a type of structure with multi-layer, which can capture highly nonlinear mappings between inputs and outputs. Over the past few years, CNNs have been succeed in solving computer vision tasks such as image classification [6], [7], object detection [8], [9], and semantic segemntation [10], [11]. This success was due to the following three main reasons: 1) CNNs model

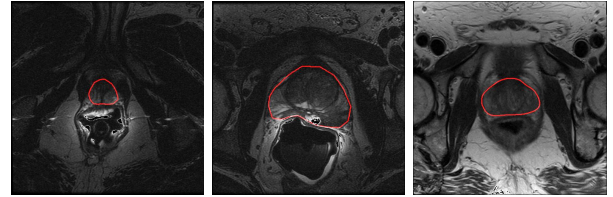


Fig. 1: Different forms of prostate MR slices with manual annotation.

has more expressive ability compared with traditional model; 2) huge amounts of annotated big datasets are available; 3) the powerful Graphics Processing Units(GPUs) accelerate the development of CNNs effectively. Particularly, CNNs have been employed successfully in the field of medical image analysis [12], [13], [14].

As for the medical image segmentation, each pixel should be labeled with high accuracy in order to assist doctors in medical treatment. Over the past few decades, some automatic or semi-automatic prostate MR segmentation algorithms have been made great progresses. For example, the approach in [15] is the first study to use edge volumes to register MR volumes for multi-atlas-based prostate segmentation. The first stage is to find a cuboid that contains the whole prostate as small cubage as possible. Secondly, the top best registered atlases are chosen for the label fusion by measuring mutual information in the vicinity of the prostate. [16] proposed a fully automated method for prostate segmentation based on context information. A volume of interest(VOI) was detected by super-voxel segmentation, and then the probability map was generated by a set of trained Random forest(RF) classifiers. [17] proposed a novel multi-feature landmark-free AAM architecture for using gradient and intensity information. This framework utilized the level-set method to locate region-of-interest. Besides, it allowed for incorporation of multiple image derived attributes(IDAs).

As we have been introduced above, CNNs have a strong representation of image features and have been applied to the medical image segmentation. For example, the representative network U-Net [18] was used to train end-to-end for segmentation of neuronal structures in electron microscopic stacks and won the ISBI cell tracking challenge 2015. The author modified and extended the FCN [10] architecture such that it works with very few training images and yields more precise

segmentations; [19] proposed a novel two-stage automated prostate segmentation model with deep network. Firstly, a set of prostate proposals in transverse plan were generated by recognizing the position of prostate region. Then, they constructed a graph to select the best prostate proposals for the final accurate prostate segmentation. A multi-scale CNNs architecture was proposed in [20] to obtain accurate brain segmentation from a single MR image modality. This network used the multiple patch sizes and multiple convolution kernel sizes to acquire multi-scale information. [21] proposed a fully automatic method for skin lesion segmentation by leveraging a 19-layer deep CNNs. They designed a novel loss function based on Jaccard distance in order to avoid the imbalance problem between foreground and background pixels. Medical organ segmentation involves assigning a label to each pixel in the slices. From the above public results, the application of CNNs in medical image segmentation were very successful. But we've been trying to find a better deep learning approach to apply to the prostate MR images segmentation task.

The HNN [22] was first proposed under the name "Holistically-nested edge detection" as an image-to-image edge prediction by means of fully CNNs and deeply-supervised nets [23]. HNN fused the extracted features from hidden layers in order to acquire accurate edge and object boundaries. This special "fusion" operation which using the extracted features of the middle layers as fully as possible has been applied to the task of object detection and segmentation successfully [24], [25]. HNN has been extended to the medical image segmentation and achieved record-breaking performances. For instance, [26] used the holistically-nested neural networks(HNN) to segment lymph node cluster. The outputs of two separated HNN models (trained on interior and boundary binary masks respectively) combined in different structured optimization methods to obtain more accurate pixel-wise label predictions. [27] presented a holistic learning method in abdominal CT scans, integrating the organ interior and boundary maps via robust spatial aggregation using random forest. The proposed approach was capable of generating boundary preserving pixel-wise class label maps for pancreas segmentation. Inspired by the previous HNN related work, we believe that HNN can achieve great improvements in prostate image segmentation. In this paper, a new framework based on HNN was proposed to segment prostate MR images automatically. We modified the original HNN via putting additional skip connections between different side outputs in order to combine multi-scale and multi-level information. Each side output layer is still as a monitor to deeply supervise the information delivery of hidden layers. All side outputs will be integrated to produce precise segmentation result in an image-to-image fashion. Compared with the original HNN, our enhanced HNN has a stronger ability of learning hierarchical feature representations from data.

The rest of this paper is organized as follows. We first describe the architecture of our network and give the formulation in Section II. The details for training our enhanced Holistically-Nested Networks is discussed in Section III. Section IV presents the experimental results and corresponding discussions. Finally, we summary this paper in Section V.

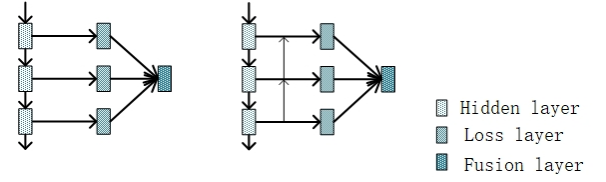


Fig. 2: Two different architectures. The left is the ordinary HNN illustration, the right model with skip connections can combine hidden information between shallower layers and deeper layers.

II. METHODS

The extracted features in CNNs are passed from shallower layers to deeper layers. we consider making full use of the intermediate information as much as possible. HNN [22] provided us an effective method for deeply supervise the performances for hidden layers in the edge detection task. As explained in [22], shallower side outputs can capture rich spatial information and deeper side outputs can better locate region-of-interest. But there are no direct connections between the side output in each stage. Besides, segmentation task is more difficult than edge detection because of the semantic information should be extracted accurately. So we try to connect multi-level information as much as possible. The skip connections are added between different separated stages for this purpose. Fig. 2 provides an illustration of the different HNN models. In this way, the high-level features are transformed from deeper layers to shallower layers. This is an effective special design for making full use of the intermediate information while combine multi-level features. Our proposed enhanced HNN framework as shown in Fig. 3. Moreover, the VGGNet [28] has demonstrated that the network depth has an crucial importance of classification accuracy. The deeper network is needed necessary for this gland segmentation task. Then we add two other convolutional layers with kernel size 3*3 in each side output layer. And we use the same bilinear interpolation operation as in HNN [22] for up-sampling.

A. HNN

The input training datasets are denoted as $Z = \{(X_n, Y_n), n = 1, \dots, N\}$, where $X_n = \{x_i^n, i = 1, \dots, |X_n|\}$ denotes the original input image and $Y_n = \{y_i^n, i = 1, \dots, |X_n|\}$, $y_i^n \in [0, 1]$ denotes the corresponding ground truth for image X_n . Each image is considered holistically and independently. We denote the collection of all standard network layer parameters as $\mathbf{W}$. There are totally M side-output layers in this network. Each side output is associated with a classifier which the corresponding weight can be denoted as $\mathbf{w} = \{\mathbf{w}^1, \mathbf{w}^2, \dots, \mathbf{w}^M\}$. Hence, the following objective function of side output can be defined as:

$$L_{side}(\mathbf{W}, \mathbf{w}) = \sum_{m=1}^M \alpha_m l_{side}^m(\mathbf{W}, \mathbf{w}^m) \quad (1)$$

where l_{side}^m denotes the image-level class-balanced cross-entropy loss function [22] for m th side outputs, α_m is the weight of the m th side loss. Finally, a weighted-fusion layer

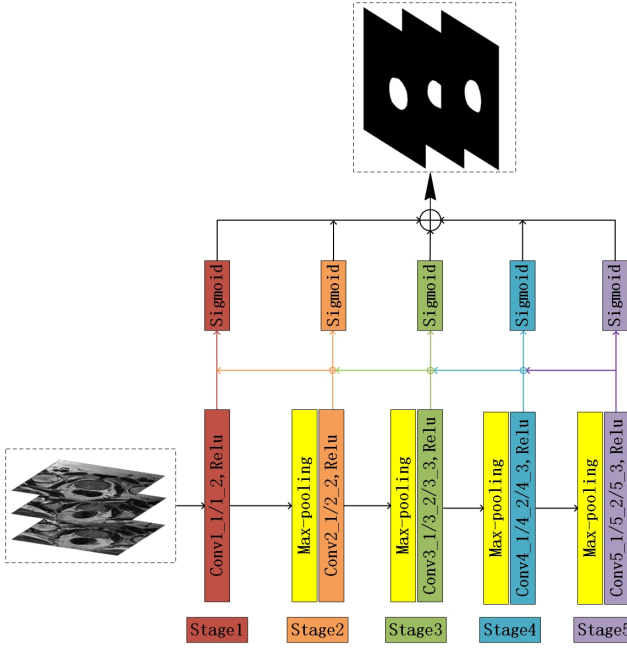


Fig. 3: Architecture of the proposed enhanced HNN. In this illustration, different color modules indicate different stages.

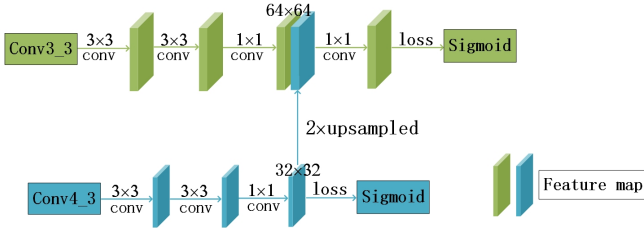


Fig. 4: An illustration of skip connection.

was added to the network that merging the information from all the side-outputs. The loss function in fusion layer can be defined as:

$$L_{fuse}(\mathbf{W}, \mathbf{w}, \mathbf{f}) = h(Y, \sigma(\sum_{m=1}^M f_m A_{side}^m)) \quad (2)$$

where $\mathbf{f} = (f_1, \dots, f_M)$ denotes the fusion weight, A_{side}^m are activations of the m th side output layer, $\sigma(\cdot)$ denotes the sigmoid function, $h(\cdot)$ denotes the distance between the fused map and the ground truth map. We also use cross-entropy loss for this purpose [22]. Hence, we consider all loss in this network, the final loss function can be given by :

$$L_{final}(\mathbf{W}, \mathbf{w}, \mathbf{f}) = L_{fuse}(\mathbf{W}, \mathbf{w}, \mathbf{f}) + L_{side}(\mathbf{W}, \mathbf{w}) \quad (3)$$

In HNN, side output layers are cascaded after the last convolutional layer of each stage. Each side output is composed of a convolutional layer with kernel size 1×1 , and followed a deconvolutional layer for up-sampling this future map.

B. Enhanced HNN

1) *Skip connection*: We also use a standard cross-entropy loss and compute the loss function over all pixels in a training image $X_n = \{x_i^n, i = 1, \dots, |X_n|\}$ and ground truth map $Y_n = \{y_i^n, i = 1, \dots, |X_n|\}$. Our loss function can be defined as follows:

$$\begin{aligned} \hat{l}_{side}^m(\mathbf{W}, \hat{\mathbf{w}}^m) = & -\beta \sum_{j \in Y_+} \log Pr(y_j = 1 | X; \mathbf{W}, \hat{\mathbf{w}}^m) \\ & - (1 - \beta) \sum_{j \in Y_-} \log Pr(y_j = 0 | X; \mathbf{W}, \hat{\mathbf{w}}^m) \end{aligned} \quad (4)$$

Here, β is simply $|Y_-|/|Y|$ and $1 - \beta = |Y_+|/|Y|$, where $|Y_-|$ and $|Y_+|$ denote the ground truth set negatives and positives, respectively. The class probability $Pr(y_j = 1 | X; \mathbf{W}, \hat{\mathbf{w}}^m) = \sigma(a_j^m) \in [0, 1]$ is computed on the activation value at each pixel j using sigmoid function $\sigma(\cdot)$.

The skip connections are used to combine different side outputs from later stages to early stages. Fig. 4 provides us a simple explanation of skip connection. Hence, the new side activation $\hat{K}_{side}^m$ at m th side output can be defined as:

$$\hat{K}_{side}^m = \begin{cases} s_i^m \hat{K}_{side}^i + \hat{A}_{side}^m, & m = 1, \dots, 4 \\ \hat{A}_{side}^m, & m = 5 \end{cases}$$

where $\hat{A}_{side}^m \equiv \{a_j^m, j = 1, \dots, |Y|\}$ are activations of the side output layer m , s_i^m denotes the weight of skip connection from i th side output to m th side output ($i = m+1$). Then, the new prostate region prediction maps $\hat{Y}_{side}^m = \sigma(\hat{K}_{side}^m)$ can be obtained at each side output layer. The new side output loss function and fusion loss function can be respectively given by:

$$\hat{L}_{side}(\mathbf{W}, \hat{\mathbf{w}}, \mathbf{s}) = \sum_{m=1}^M \alpha_m \hat{l}_{side}^m(\mathbf{W}, \hat{\mathbf{w}}^m, \mathbf{s}) \quad (5)$$

and

$$\hat{L}_{fuse}(\mathbf{W}, \hat{\mathbf{w}}, \mathbf{f}, \mathbf{s}) = \hat{h}(Y, \sigma(\sum_{m=1}^M f_m \hat{K}_{side}^m)) \quad (6)$$

where $\mathbf{s} = \{s_i^m\} (i = m+1)$, $\hat{h}$ denotes the distance between the new fused prediction maps and ground truth maps. Thus, the new final loss function can be written by:

$$\hat{L}_{final}(\mathbf{W}, \hat{\mathbf{w}}, \mathbf{f}, \mathbf{s}) = \hat{L}_{fuse}(\mathbf{W}, \hat{\mathbf{w}}, \mathbf{f}, \mathbf{s}) + \hat{L}_{side}(\mathbf{W}, \hat{\mathbf{w}}, \mathbf{s}) \quad (7)$$

we also minimize the final loss function via standard stochastic gradient descent followed [22].

2) *System overview*: Considering entire network architecture in Fig. 3, each side output is independent and linked together. The key characteristic of our proposed network is that each side output layer plays an important role in supervising the hidden layers. Besides, high-level features can be passed from later stages to early stages due to the additional skip connections. The shallower layers can learn richer low-level features to better refine the prediction maps from deeper side

output layers. And then we will obtain more accurate prostate region prediction in later stages. In addition, we put additional two other convolutional layers with kernel size 3*3 to expand the side output layers and increased the depth of the network, which is more effective for learning the features of the data. The parameters in output layer can be updated by error back-propagation in weighted-fusion layer (Equation 6). In this architecture, the enhanced HNN can obtain multi-scale feature maps at each stage. The final unified output is a weighted average of all prediction maps:

$$\hat{Y}_{final} = Average(\hat{Y}_{fuse}, \hat{Y}_{side}^1, \dots, \hat{Y}_{side}^M) \quad (8)$$

III. TRAINING AND EXPERIMENT

A. Dataset and pre-processing

The public prostate dataset MICCAI2012 [29] consists of transversal T2-weighted MR images. The MR scans in this dataset which were acquired by different clinical settings, showing obvious variations in dynamic range, voxel size and appearance. We evaluated our enhanced HNN on the all 50 training patient samples of the dataset. In terms of the medical image analysis, training such a network may cause the problem of overfitting because of the limited number of available medical images. To increase the robustness and reduce the phenomenon of overfitting, we utilized the strategy of data augmentation to enlarge the original prostate dataset. The methods were used for data augmentation include rotation, zoom, mirror and translation operations.

However, there is an unbalanced problem for the prostate MR images, which the region-of-interest occupies a small area, and the background area accounts for a large proportion of the total slices. In order to speed up the convergence of network and reduce the negative samples, we remove the unnecessary region which do not contain prostate pixels and crop the original slices to a smaller region with 256*256 size. Moreover, the weak edge boundaries between prostate tissue and background are difficult to be distinguished. So we enhanced the contrast in edge boundaries for the MR slices using a diffusion method based on [30]. After the image pre-processing step, we mixed all slices together and extended the data to around 8500 MR slices.

B. Implementation details

Our proposed network was based on the public available deep learning library Caffe [31] and the open implementation of [22]. In the training phase, we choose HNN [22] as our pre-trained Caffe model for initializing network parameters. The initialized learning rate and momentum are set to 1e-6 (multiplied by 10 every 10000 iterations) and 0.9, respectively. All experiments were implemented in a single Nvidia GTX TITAN X GPU with 12G memory and CUDA edition is 7.5.

IV. RESULT AND DISCUSSION

A. Qualitative performance evaluation

In order to verify the effectiveness of the enhanced HNN, we compared it with FCN [10], U-Net [18] and HNN [22]. We use the same dataset to train and test the other three

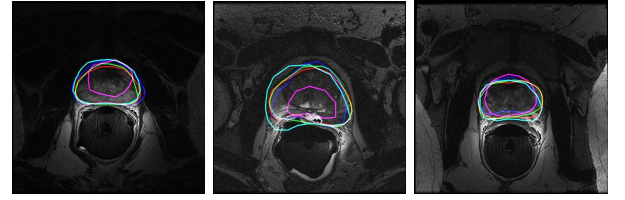


Fig. 5: The segmented prostate MR slices with final contour annotation results using four different methods. Red: manual annotation; Green: enhanced HNN; Blue: HNN; Cyan: U-Net; Magenta: FCN.

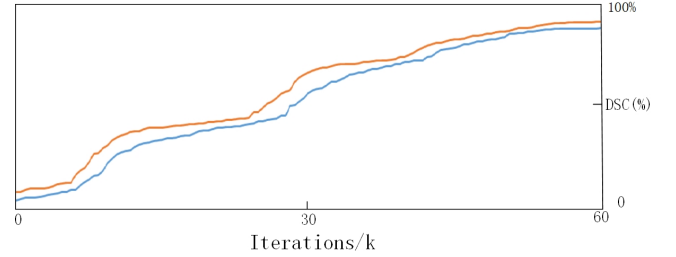


Fig. 6: The DSC(%) value for HNN and Enhanced HNN increase with the iterations respectively(Orange: enhanced HNN; Blue: original HNN).

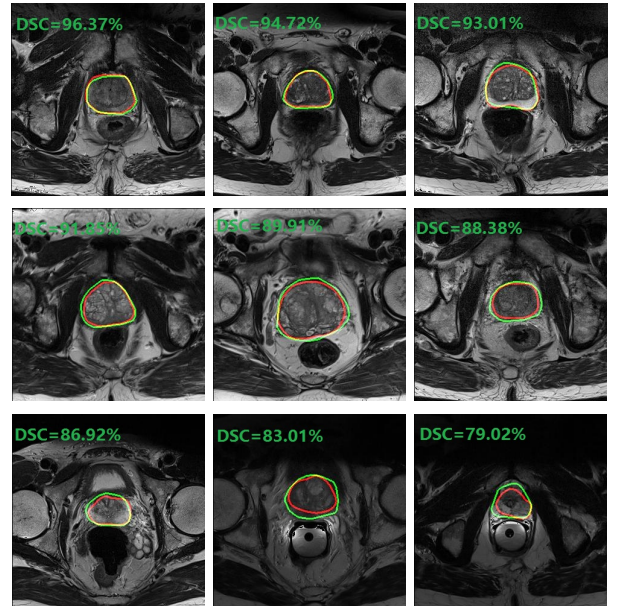


Fig. 8: The final segmented prostate MR slices with manual annotation(Red) and enhanced HNN(Green) respectively(Due to the overlap of red and green annotations, a yellow outline will be displayed).

methods. Fig. 5 shows us some segmented prostate MR slices with different methods. As can be seen from the picture, the enhanced HNN can better segment the deformed shapes of prostate than original HNN. Although the encoder-decoder architecture U-Net can roughly describe the shape of prostate

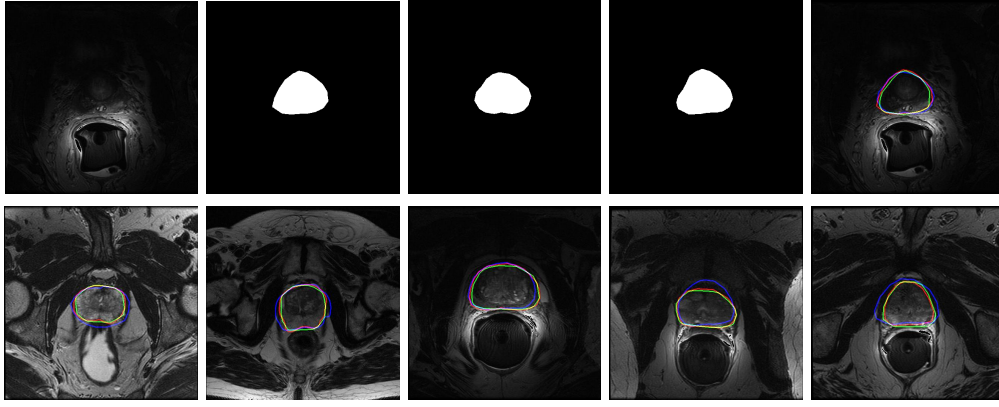


Fig. 7: Final segmentation results comparison between HNN and Enhanced HNN. First line(from left to right): original slice, ground truth, HNN result, Enhanced HNN result, final contour annotation; Second line: five final contour annotated slices (Red: manual annotation; Blue: HNN; Green: enhanced HNN)

TABLE I: Quantitative comparison with other segmentation methods in DSC(%) and HD(mm).

Methods	DSC(%)				HD(mm)			
	Minimum	Maximum	Median	Mean	Minimum	Maximum	Median	Mean
FCN [10]	57.84	91.60	82.33	77.64	8.19	85.36	17.33	20.18
U-Net [18]	71.57	94.51	85.76	83.47	5.25	61.04	12.51	14.46
HNN [22]	68.88	93.76	88.15	85.53	5.68	72.49	14.73	15.39
Enhanced HNN	76.31	96.20	90.11	87.61	4.19	65.37	12.44	12.98

area, it is worse than HNN and enhanced HNN. The FCN method can not find the correct location of the prostate region, it is the worst algorithm in the four approaches.

B. Quantitative performance evaluation

In order to evaluate the segmentation accuracy between our segmentation result and the manual ground truth, we utilized the Dice Similarity Coefficient(DSC) and Hausdorff Distance(HD) as our quantitative metrics for evaluation [29]. These equations are given by:

$$DSC(A, B) = \frac{2|A \cap B|}{|A| + |B|} \quad (9)$$

$$HD(A_s, B_s) = \max(h(A_s, B_s), h(B_s, A_s)) \quad (10)$$

where $|A|$ and $|B|$ in equation 9 denote the region of the algorithm segmentation and reference segmentation respectively. In equation 10, A_s and B_s are the surface points of the algorithm segmentation and reference segmentation respectively. Besides, $h(A_s, B_s) = \max_{a_i \in A_s} \min_{b_j \in B_s} \|a_i - b_j\|$, $h(B_s, A_s) = \max_{b_j \in B_s} \min_{a_i \in A_s} \|b_j - a_i\|$. $\|\cdot\|$ is the Euclidean distance. The equation of DSC essentially measures the overlap between the segmentation result and ground truth. HD value measures the maximum degree of mismatch between two point sets of prostate contour regions.

TABLE I provides us the statistical results. The mirrored version structure U-Net elevated the mean DSC from 77.64% to 83.47%, which is a significant improvement over the FCN

architecture. Then, the feature fusion method HNN raised this metric about 2.06%. Our enhanced HNN model further improved the segmentation performance and obtained the highest DSC value 87.61%. Fig. 6 also shows the increasing trend of the DSC value with the number of iterations between original HNN and Enhanced HNN. As can be seen from the Fig. 6, the accuracy of the Enhanced HNN on the validation set has a certain improvement over the original HNN model. In addition, it also obtained the highest value in the three other aspects of DSC. The average HD values of our method, HNN, U-Net and FCN are 12.98mm, 15.39mm, 14.46mm and 20.18mm respectively. Our enhanced HNN still obtained the best results in mean HD metric.

C. Discussion

From the comparisons in TABLE I and Fig. 5, it is obvious that the FCN obtained the worst result between the four different methods, the results obtained from U-Net and HNN are close to each other. Our enhanced HNN obtained best results and showed stronger robustness than the original HNN and two other algorithms. Fig. 7 gives some direct comparisons of the final segmentation results between HNN and Enhanced HNN. It can be seen from the visualizations that our Enhanced HNN has a certain improvements on original HNN. Fig. 8 also shows us some different patient samples with final contour annotation. The red color denotes the manual annotation and green color denotes the segmentation results using enhanced HNN. The slices in first two lines show the successful segmentation results and three not good segmentation results were given in last three slices. In terms

of the slices with distorted shapes, our segmentation results still need to be further improved.

V. CONCLUSION

In this paper, we proposed an enhanced Holistically-Nested Networks for automatic prostate MR images segmentation. The original HNN provided an effective method for multi-level features fusion. In order to fully utilize the intermediate information, and combine the high-level and low-level features in different stages, we put additional skip connections between early and later stages instead of merely integrating all the side outputs into the fusion layer. The enhanced HNN framework can merge multi-level and multi-scale features via holistically-nested learning, and immediately produces fused outputs in an image-to-image manner. Experimental results have demonstrated that this proposed architecture improved the accuracy of prostate MR images segmentation.

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